

Computer Organization and Architecture · 21CSS201T

Unit 1 · Session 2 · SLO 2 — Solved Worksheet

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Unit – 1 : Session – 2 : SLO – 1 | Digital systems and number systems

■ MCQs

1. c — Combinational
2. c — Discrete physical quantities
3. d — Hexadecimal
4. b — Synchronous
5. c/d — 4 bits (rounded up from $\log_2 10$)

■ Two-Mark Questions

1. Synchronous sequential systems update their state only at fixed clock intervals (clock-driven, easier to design and debug); asynchronous systems change state whenever inputs change (no clock, faster but prone to race conditions and harder to analyze).
2. A binary signal is a signal that can take only one of two discrete values (0 or 1). Physical examples: voltage levels 0 V / +5 V in a circuit; switch OFF/ON; magnetic polarity on a storage disk.

■ Scenario-Based Long Answer

- Suitable system: a synchronous SEQUENTIAL digital system (a counter built from flip-flops).
- State: the current count is stored in flip-flops (internal state with memory). Input: count pulses from the wheel sensor and a reset line. Output: the state drives the odometer display.
- Synchronous is justified because all flip-flops update together on a common clock edge, so the count changes cleanly and predictably with no race conditions; an asynchronous ripple counter could show transient wrong values. A synchronous counter with a clear/reset input meets the precise-tracking and reset requirements.

■ Match the Following

- Binary — b) Base 2
- Hexadecimal — a) Base 16
- Byte — c) 8 bits
- Synchronous System — d) Updates at fixed intervals
- Asynchronous System — e) Updates at any time

■ Fill in the Blanks

1. combinational 2. octal 3. nibble 4. 0 and 1 5. Digital

■ Scenario-Based Small Activity

- A basic calculator uses BOTH: combinational logic (adders/ALU that compute results from current inputs) and sequential logic (registers/memory that hold operands and running totals).
- Components: (i) Inputs — keypad; (ii) Outputs — display; (iii) Memory/state — registers storing the first operand and intermediate result.

- Classification logic: the arithmetic unit alone is combinational (output depends only on present inputs), but storing numbers between key presses requires state, making the overall calculator a sequential system.

■ True or False

1. FALSE — A nibble is 4 bits.
2. FALSE — A digital signal takes only discrete values (0 or 1).
3. TRUE
4. FALSE — Sequential systems are usually synchronous.
5. TRUE